View Reviews

Paper ID
2752

Paper Title
Declarative Explanations for Graph Neural Networks: A Demonstration

Track Name
Demo

Reviewer #1

Questions

1. Overall Rating
Reject

2. The architecture and functionalities of the tool/software are clearly articulated.
Partially

3. The demonstration scenario is clearly articulated
No

4. Please rate the potential degree of audience interaction of the demo scenario
Medium

6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.
(S1) Explainability for DL, especially GNNs, is an important topic.

(S2) The system demonstrated seems sophisticated, and the research is backed up by a full research paper

(S3) The source code is publicly available

7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.
(W1) The Demonstration is difficult to follow, making it difficult to evaluate how helpful SliceGXQ is in realistic scenarios

8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.
(W1) The Demonstration is difficult to follow, making it difficult to evaluate how helpful SliceGXQ is in realistic scenarios

Video
- The video uses an auto-generated monotonic voice, which makes it hard to follow and

undermines the professionalism of the presentation. The video also exceeds the 5 min time limit by almost 2 min.

Section 2
- The components introduced at the beginning of Section 2 do not directly map to the components shown in Figure 1. This makes it difficult to follow the text and to mentally map the description to the shown workflow.

Section 4
- General issue (applies to all scenarios): None of the scenarios follow a clear narrative structure. Each scenario should explicitly state: (i) the dataset and its context, (ii) the analysis question or problem being addressed, and (iii) a step-by-step walkthrough of how SliceGXQ is used and how its outputs are interpreted. Currently, all four scenarios lack this structure to varying degrees.
- Figure 4 does not correspond to the UI shown in the accompanying video, and there is no intuitive mapping between the two. This makes both the video and the written demo description difficult to follow. The figure should either reflect the actual UI or be accompanied by a clear explanation of the correspondence.
- Scenario 2: The scenario begins with a specific misclassified node without providing any dataset context or motivating the example. The reader cannot assess whether this is a realistic or representative situation. Background on the dataset and the relevance of the chosen node should be provided before diving into specifics.
- Scenario 2: The claim that "explanations of layers 1 to 3 reveal that M performs correctly up to layer 2, suggesting that the misclassification likely arises at hop 3" is not sufficiently justified. While Figure markers indicate errors, neither the text nor the figure explains why this conclusion follows. The reasoning should be made explicit.
- Scenario 3: The "coauthors" and "amazon" datasets are used without introduction and their content and relevance to the demonstrated analysis are never explained. Additionally, reporting exact node counts adds little value and could be omitted. More importantly, this scenario reads as a scalability evaluation rather than a usage scenario; it should either be reframed as such or replaced with a genuine end-to-end use case.
- Scenario 4: The YelpChi dataset is not introduced. As with the other scenarios, a brief description of the dataset and the concrete analysis question should precede the walkthrough.

Detailed Comments:

- Specific examples of SPARQL-like queries are shown in Figure 2, but they are never discussed or used to demonstrate the system.
- The text runs over at the beginning of Sec 2.
- Fig 5 font size for plot too small.

Reviewer #2

Questions

1. Overall Rating
Weak Accept

2. The architecture and functionalities of the tool/software are clearly articulated.
Yes

3. The demonstration scenario is clearly articulated
Yes

4. Please rate the potential degree of audience interaction of the demo scenario
Medium

6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.
S1. The paper addresses an important problem in GNN explainability, especially the limitation that many existing explainers only provide one-shot explanations for final predictions.

S2. The SPARQL-like query interface is a useful system contribution, allowing users to specify target nodes, layers, constraints, and explanation modes in a flexible and declarative way.

S3. The demo workflow is clear and practical, covering dataset/model selection, query-based explanation, visualization, model diagnosis, and debugging.

7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.
W1. The novelty boundary between this demo system and the authors' prior SliceGX work is not fully clear.

W2. The evaluation and demo evidence are somewhat limited, especially regarding scalability, runtime, and interactive usability.

W3. The query language and explanation workflow would benefit from clearer semantics and more concrete examples.

8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.
D1. The paper states that layer-wise GNN explanations are developed in the full SliceGX paper, while this demo builds SliceGXQ around them. However, it could better clarify what is new in this demo, such as the declarative interface, visualization system, system integration, or scalability support.

D2. The paper includes useful scenarios such as model diagnosis, large-scale interpretation, and debugging, but the empirical discussion is brief. More details about runtime settings, query complexity, explanation size, and system responsiveness would make the demo more convincing.

D3. The query interface is appealing, but its supported patterns, error handling, and usability are under-explained. More examples showing different queries and their corresponding outputs would help users understand how expressive and practical SliceGXQ is.

Reviewer #3

Questions

1. Overall Rating
Weak Accept

2. The architecture and functionalities of the tool/software are clearly articulated.
Yes

3. The demonstration scenario is clearly articulated
Yes

4. Please rate the potential degree of audience interaction of the demo scenario
High

6. At most three strong points about the proposal. Please be precise; clearly explain the value and nature of the contribution.
S1. The tool tries to address the complex "black-box" nature of layer-wise GNN message passing over substantial graphical architectures by providing a declarative and interactive generation and access of layer-wise explanations for GNNs over large-scale
graphs. This is very helpful to understand the inner situation when the GNNs are in the designing or profiling stage

S2.It delivers progressive, layer-by-layer visibility into deep model inference.

S3. The demo is interesting; it shows solid performance and scaling traits across large graph footprints, including the massive Amazon data topology

7. At most three weak points about the proposal. Please be precise here. The details can be in the relevant field below.
W1. The sub-graph calculation relies on a localized greedy heuristic that remains vulnerable to missing globally optimal explanation nodes.
W2. The layer fine-tuning and structural pruning still require manual designer invocation.

8. Elaborate on the above fields. Please number each point and provide as constructive feedback as possible.

In general, the demo is interesting and useful. Some of the designs need to be refined to make it a really useful tool (W1 and W2).